

Question No:4

Question description

The students of college *XYZ* are getting jealous of the students of college *ABC*. *ABC* managed to beat *XYZ* in all the sports and games events. The main strength of the students of *ABC* is their unity. The students of *XYZ* decide to destroy this unity. The geeks of *XYZ* prepared a special kind of perfume. Anyone who inhales this perfume becomes extremely violent. The students of *XYZ* somehow manage to spread this perfume throughout *ABC*'s campus atmosphere.

There are *N* boys $(1, 2, 3, \dots, N)$ and *N* girls $(1, 2, 3, \dots, N)$ in *ABC* college. Each boy has a crush on a single girl and each girl has a crush on a single boy. Since the perfume has been inhaled by each and every student of *ABC* college, every student decides to beat up his/her crush's crush, ie. , if boy *x* has a crush on girl *y* and girl *y* has a crush on boy *z*, *x* will beat *z* up, provided, of course, if *x* and *z* is not the same person.

The doctor of *ABC* college foresees this situation. He cannot stop so many people from beating each other up, however, he can be prepared for the worst-case patient(s). The worst-case patient(s) will be the patient(s) who get(s) beaten up by the maximum number of students. The doctor comes to you for help. He has 2 questions for you :

- 1. What is the number of beatings received by the worst-case patient(s) ?
- 2. What is the total number of pairs of students who ended up beating up each other ?

Constraints :

$$1 \leq T \leq 10$$

$$1 \leq N \leq 10^5$$

Input :

The first line comprises of *T*, the number of test cases. Each test case comprises of 3 lines. The first line consists of *N*.

The next line consists of *N* space separated natural numbers between 1 and *N* inclusive such that the *i*th number denotes the the crush of boy *i*.

The next line consists of *N* space separated natural numbers between 1 and *N* inclusive such that the *i*th number denotes the the crush of girl *i*.

Output :

For every test case, on a new line, print two space separated integers, the answer to doctor's question 1 followed by answer to doctor's question 2.

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
2
3
2 2 1
3 2 1
4
2 3 4 1
2 3 4 1
```

EXPECTED OUTPUT

```
1 0
1 4
```

Test Case 2

INPUT (STDIN)

```
2
4
4 2 1 3
4 2 1 2
3
2 4 1
2 4 1
```

EXPECTED OUTPUT

```
1 0
1 1
```

Code:

```
1  #include <stdio.h>
2  #include <stdlib.h>
3
4  int main() {
5      int T;
6      scanf("%d", &T);
7
8      while (T--) {
9          int N;
10         scanf("%d", &N);
11
12         int *a = malloc((N + 1) * sizeof(int));
13         int *b = malloc((N + 1) * sizeof(int));
14
15         int *boyTarget = malloc((N + 1) * sizeof(int));
16         int *girlTarget = malloc((N + 1) * sizeof(int));
17
18         int *boyCount = calloc(N + 1, sizeof(int));
19         int *girlCount = calloc(N + 1, sizeof(int));
20
21         for (int i = 1; i <= N; i++) {
22             scanf("%d", &a[i]);
23         }
24
25         for (int i = 1; i <= N; i++) {
26             scanf("%d", &b[i]);
27         }
28
29         // Find the boy each boy beats up.
```

```
30     for (int i = 1; i <= N; i++) {
31         boyTarget[i] = b[a[i]];
32
33         if (boyTarget[i] != i) {
34             boyCount[boyTarget[i]]++;
35         }
36     }
37
38     // Find the girl each girl beats up.
39     for (int i = 1; i <= N; i++) {
40         girlTarget[i] = a[b[i]];
41
42         if (girlTarget[i] != i) {
43             girlCount[girlTarget[i]]++;
44         }
45     }
46
47     // Find the maximum number of beatings received.
48     int maxBeatings = 0;
49
50     for (int i = 1; i <= N; i++) {
51         if (boyCount[i] > maxBeatings) {
52             maxBeatings = boyCount[i];
53         }
54
55         if (girlCount[i] > maxBeatings) {
56             maxBeatings = girlCount[i];
57         }
58     }
```

```

58     }
59
60     // Count reciprocal pairs, once per pair.
61     int pairs = 0;
62
63     for (int i = 1; i <= N; i++) {
64         int j = boyTarget[i];
65
66         if (j != i && boyTarget[j] == i && i < j) {
67             pairs++;
68         }
69
70         j = girlTarget[i];
71
72         if (j != i && girlTarget[j] == i && i < j) {
73             pairs++;
74         }
75     }
76
77     printf("%d %d\n", maxBeatings, pairs);
78
79     free(a);
80     free(b);
81     free(boyTarget);
82         free(girlTarget);
83         free(boyCount);
84         free(girlCount);
85     }
86
87     return 0;
88 }

```

Output:

Test Case-1:

```
3seem\DSA\DSA Practice\hashing\hashing.ps
2
3
2 2 1
3 2 1
1 0
4
2 3 4 1
2 3 4 1
1 4
PS C:\Users\kesha\OneDrive\Documents\3seem\DSA\DSA Practice\hashing\hashing.ps
```

Test Case-2:

```
3seem\DSA\DSA Practice\hashing\hashing.ps
2
4
4 2 1 3
4 2 1 2
1 0
3
2 4 1
2 4 1
PS C:\Users\kesha\OneDrive\Documents\3seem\DSA\DSA Practice\hashing\hashing.ps
PS C:\Users\kesha\OneDrive\Documents\3seem\DSA\DSA Practice\hashing\hashing.ps
```

Keshav Kumar Rauniyar

CH.SC.U4CSE25236